Simplification of Boolean function

- Simplification of Boolean functions is mainly used to reduce the gate count of a design.
- Less number of gates means less power consumption, cost also comes down.
- Map is a diagram made up of squares. Each square represents one minterm or maxterm.

Karnaugh Maps

- Karnaugh maps provide a systematic method to obtain simplified sum-of-products (SoPs) or product-of-sum (PoSs) Boolean expressions.
- It is ideally suited for four or less variables
- Each square represents either a minterm or maxterm. A K-map of n variables will have 2^n squares.

Minterms and Maxterms

- **Minterms:**
- Lets consider two variable x and y, there are 2^n=4 possible combinations. x'y' , x'y , xy' and xy. All are AND terms.
- Each terms is called minterm or standard product.
- **Maxterms:**
- Similarly, OR terms are called the maxterms or standard sum.

- Minterms and Maxterms for three binary variables:

			Minterms		Maxterms	
x	y	z	Terms	Designation	Term	Designation
0	0	0	x'y'z'	m_0	x+y+z	M_0
0	0	1	x'y'z	m_1	x+y+z'	M_1
0	1	0	x'yz'	m_2	x+y'+z	M_2
0	1	1	x'yz	m_3	x+y'+z'	M_3
1	0	0	xy'z'	m_4	x'+y+z	M_4
1	0	1	xy'z	m_5	x'+y+z'	M_5
1	1	0	xyz'	m_6	x'+y'+z	M_6
1	1	1	xyz	m_7	x'+y'+z'	M_7

x	y	z	F_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

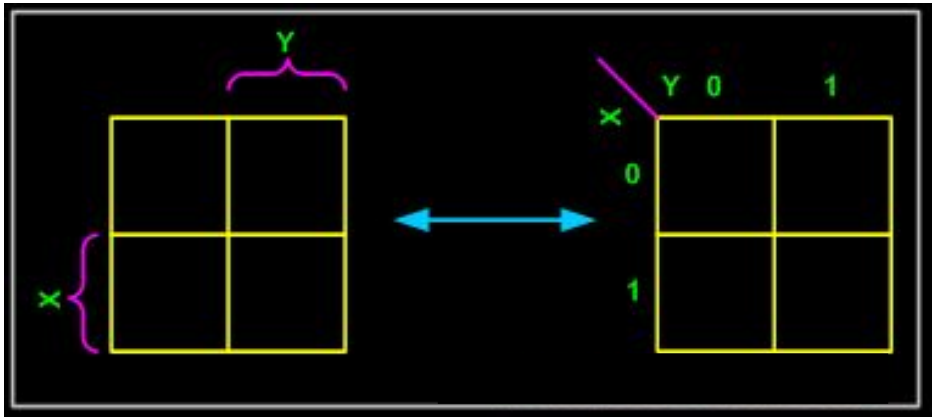
- $F_1 = x'y'z + xy'z' + xyz$ (considering 1)
- $= m_1 + m_4 + m_7$
- This is called sum of minterms or sum of product.

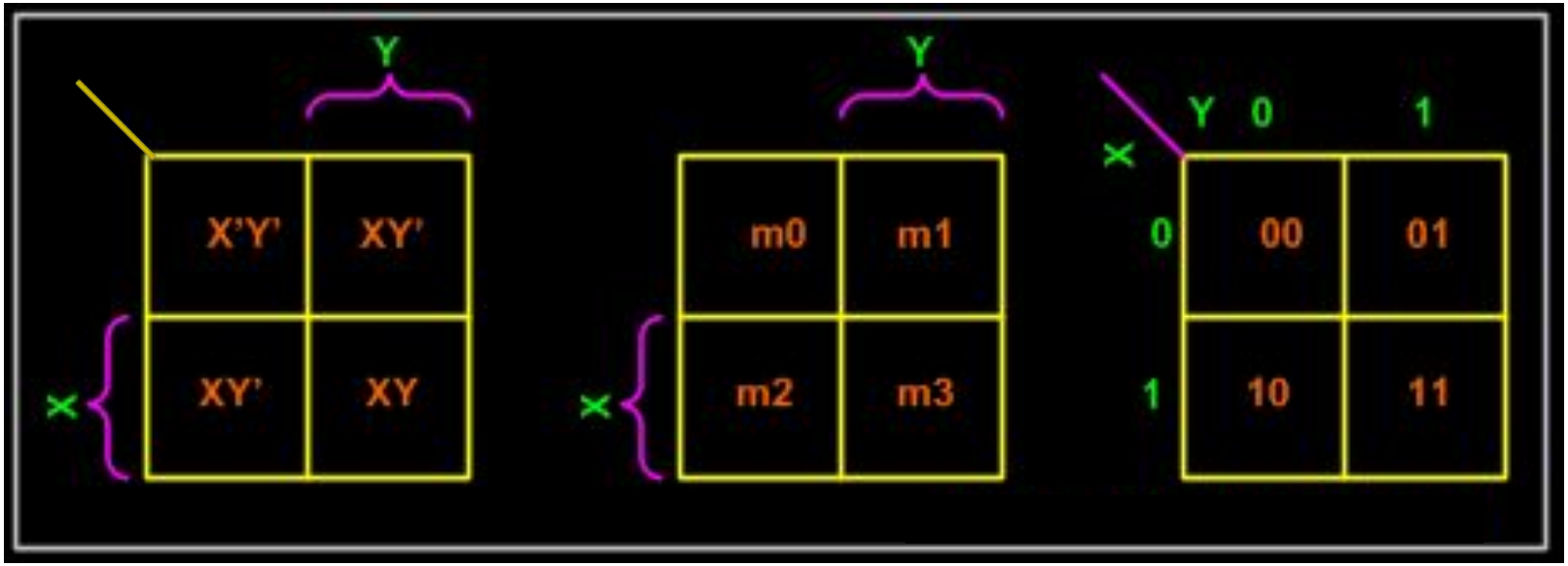
- However
- $F'_{1} = x'y'z' + x'yz' + x'yz + xy'z + xyz'$ (considering 0)
- If we take the complement of F'_1, we obtain the F_1
- $F_1 = (x+y+z)(x+y'+z)(x+y'+z')(x'+y+z')(x'+y'+z)$
- $= M_0.M_2.M_3.M_5.M_6$
- This is called product of maxterms or product of sum

2-Variable K-Map

Two variable map is the one which has got only **two variables** as input.

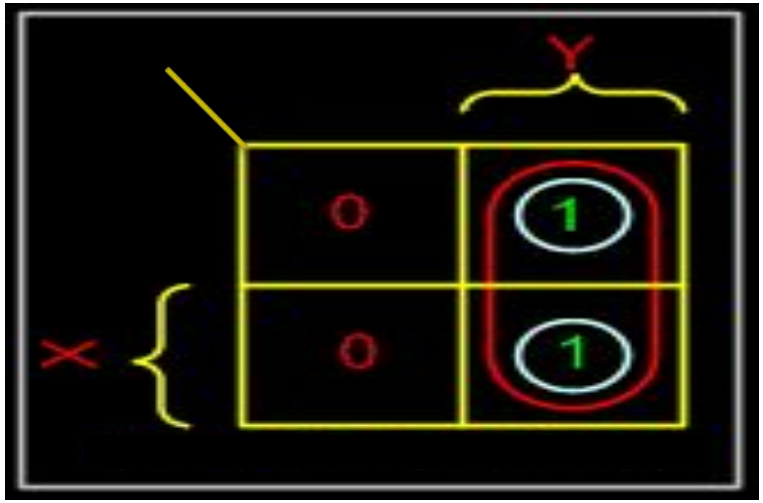
2^n squares





Example : X'Y+XY

X	Y	Output
0	0	0
0	1	1
1	0	0
1	1	1

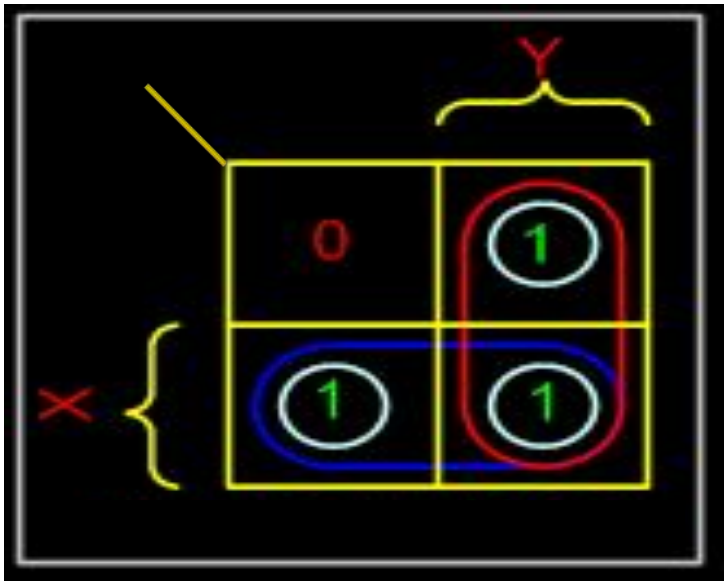


F = Y

- Grouping the terms to form single terms.
- A square containing 1 can be included in as many groups as desired.
- A group must be as large as possible.

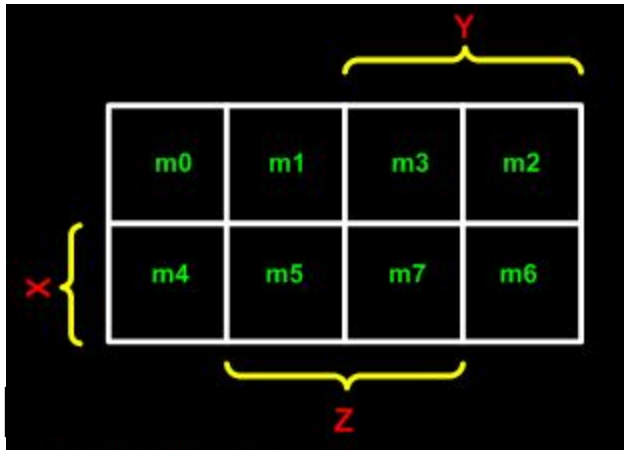
Example - X'Y+XY+XY'

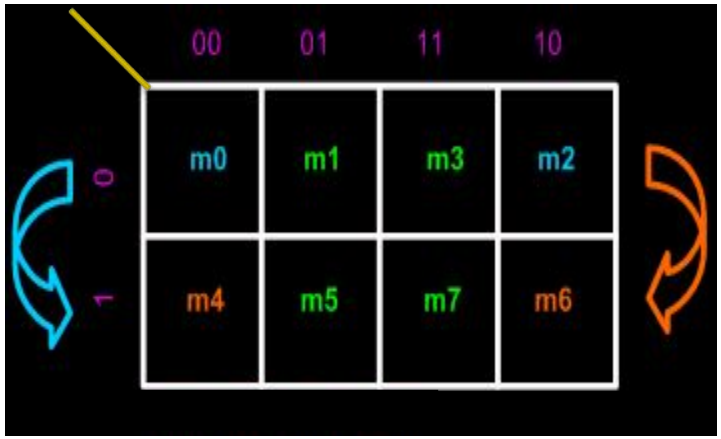
X	Y	Output
0	0	0
0	1	1
1	0	1
1	1	1



F = X + Y

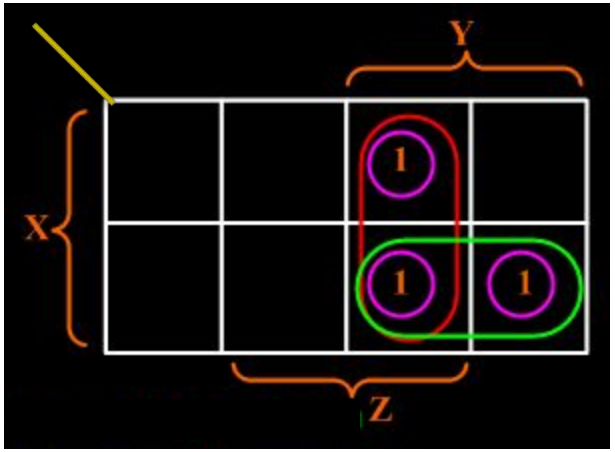
3-Variable K-Map





Example

F = XYZ'+XYZ+X'YZ

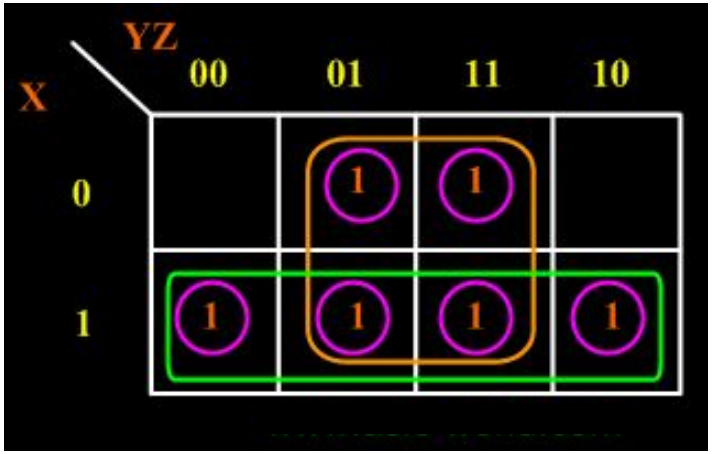


F = XY + YZ

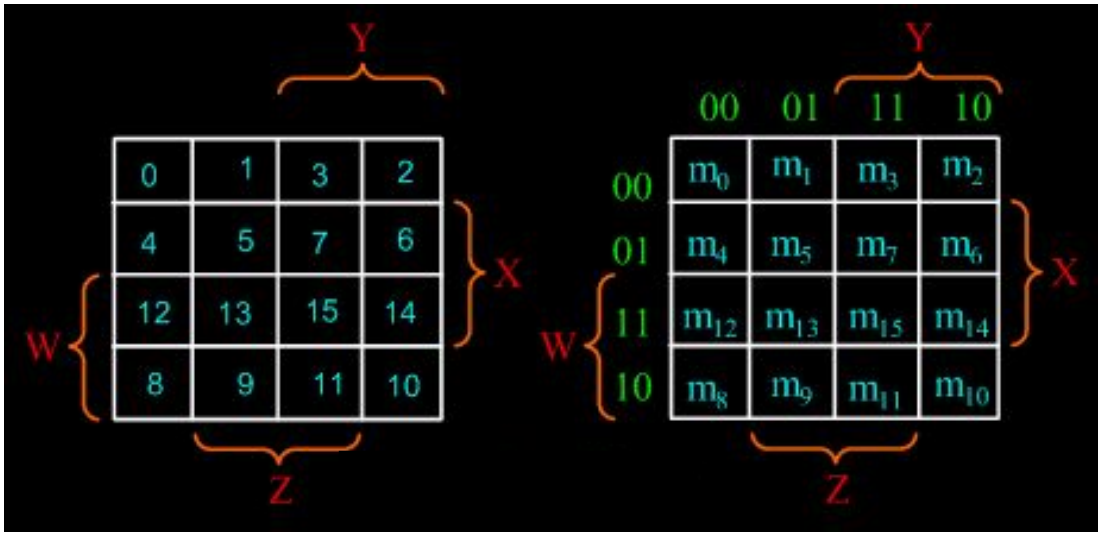
Example

$F(X,Y,Z) = \sum(1,3,4,5,6,7)$

$F = X + Z$

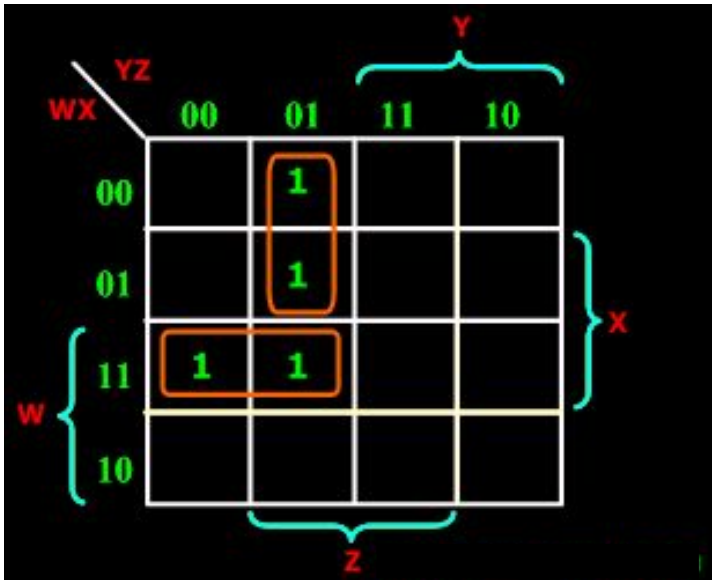


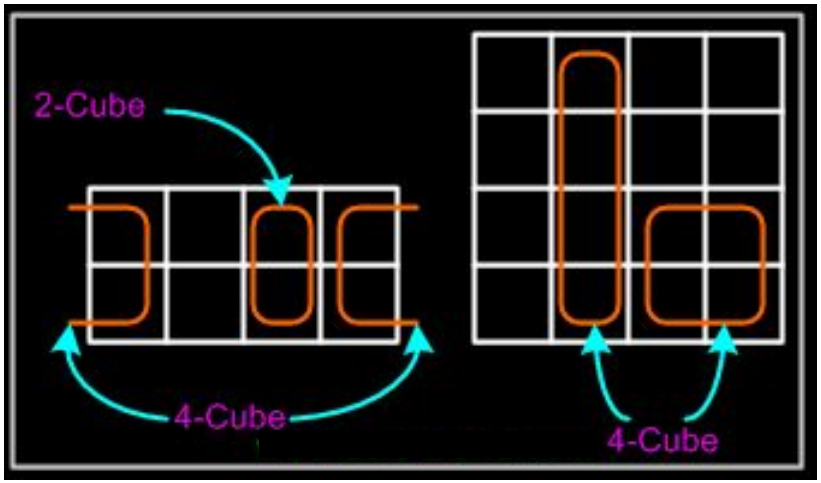
4-Variable K-Map

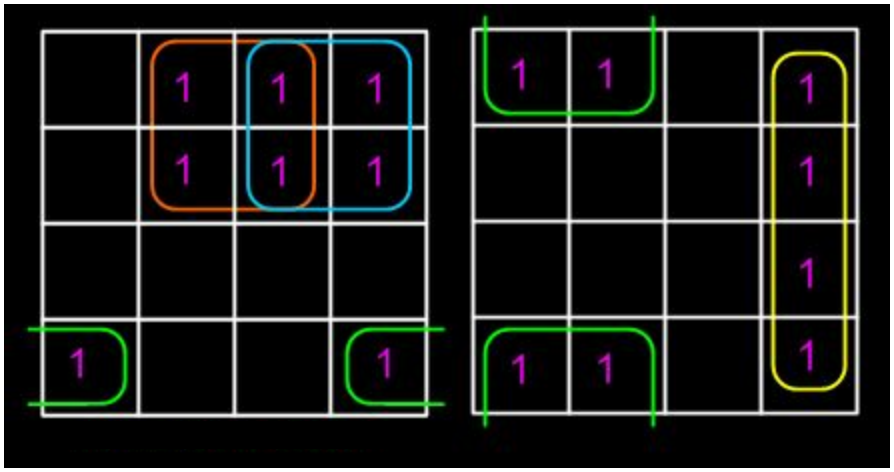


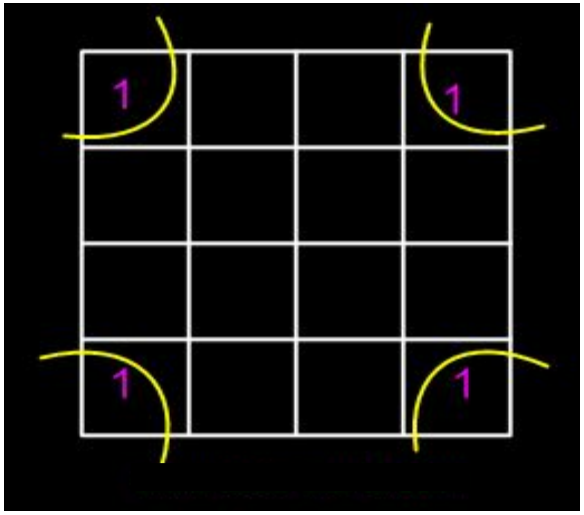
$F(W,X,Y,Z) = (1,5,12,13)$

$F = WXY' + W'Y'Z$

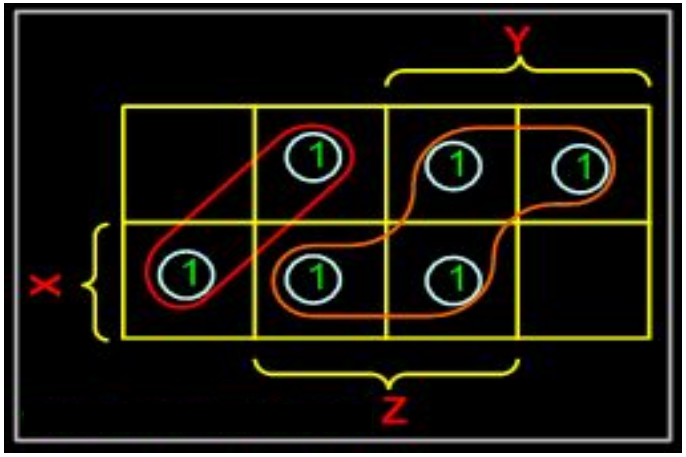








Example of invalid groups



Do exercise:

Example: 3.1, 3.2, 3.3, 3.4, 3.5, 3.6 (P. 74-80) – M.M.Mano